

General Notes

- You will need to submit two files: a zip file and an output file (either HTML or PDF). The zip file must contain at least two files: an R script and an RMD file. All files must follow the specified naming conventions, including capitalization and underscores. Replace 123456789 with your nine-digit Bruin ID in all filenames. The files to be submitted are:
 1. *123456789_stats102a_hw2.html/pdf*: An output file, either a PDF or an HTML file depending on the output you choose to generate.
 2. *123456789_stats102a_hw2.zip*: Place all your R script, RMD, and image files into a single folder named *123456789_stats102a_hw1*, then compress the folder into *123456789_stats102a_hw1.zip*.
 - *123456789_stats102a_hw2.R*: An R script file containing all the functions you wrote for the homework. In your RMD file, the first line after loading libraries should source this script file.
 - *123456789_stats102a_hw2.Rmd*: A markdown file generates the output file of your submission.
 - *included image files*: You may name these files as you choose, but you must include all image files you generated for your RMD file; otherwise, it will not knit.

If you fail to submit any of the required files you will receive **ZERO** points for the assignment. If you submit any files which do not conform to the specified naming convention, you will receive **half credit** for the assignment.

- **Your RMD file must be knittable. If your RMD file does not knit in graders' computer, you will receive at most half credit for the assignment.**
The two most common reasons RMD files fail to knit are workspace/directory structure issues and missing files. To address the first, ensure all file paths in your document are relative paths pointing to the current working directory. To resolve the second, upload all files you source or include in your RMD file to the same folder.
- Your coding should adhere to the tidyverse style guide: <https://style.tidyverse.org/>.
- All flowcharts should be done on separate sheets of paper, but be included, inline as images, in your final markdown document.
- Any functions/classes you write should have the corresponding comments as the following format.

```
my_function = function(x, y, ...){  
  #A short description of the function  
  #Args:  
  #x: Variable type and dimension  
  #y: Variable type and dimension  
  #Return:  
  #Variable type and dimension  
  Your code begin here  
}
```

NOTE: *Everything* you need to do this assignment is here, in your class notes, or was covered in discussion or lecture.

- **DO NOT** look for solutions online.
- **DO NOT** collaborate with anyone inside (or outside) of this class.
- **DO NOT** print out the whole dataset in the markdown.
- Work **INDEPENDENTLY** on this assignment.
- **EVERYTHING** you submit **MUST** be 100% your, original, work product. Any student suspected of plagiarizing, in whole or in part, any portion of this assignment, will be **immediately** referred to the Dean of Student's office without warning.

1: Teacher's Gradebook

The formatting takes 15% of this homework grade. The **efficiency**, **accuracy**, and **robustness** of your function account for 60%. The total is 75% for this question. If your functions violent any instructions of this assignment, the function will not be graded.

Requirements:

1. Draw flowcharts (or an algorithms) of the main functions sufficiently complete, clear, and concise enough to enable a person to accurately implement the functions in any programming languages.
2. Write the function(s) which accurately implements the algorithm(s) as described or requested.
3. Include the error-handling to ensure your function(s) work properly.

Note: The requirements apply to your **messy_impute()** and **tidy_impute()** functions.

One scenario that naturally creates non-tidy data is a teacher's gradebook. Table 1 provides an example with five homework assignments and seven quizzes. In this table, each row represents a student, and each column corresponds to a different assignment. While the teacher may have a gradebook with a fixed number of assignments, the number of students can vary.

Additionally, the gradebook may contain missing values due to various reasons. To analyze the data effectively, the teacher needs to impute these missing values. Imputation is the process of replacing missing values with estimated values. The simplest (though far from ideal) method is to replace missing values with the most typical value, such as the mean or median.

For this assignment, you will write two functions to impute missing values in a teacher's gradebook. Since we will retain all grades as integers, please use the floor function to round up the imputed values.

Table 1: An example of a teacher's gradebook

UID	Homework1	Homework2	...	Homework5	Quiz1	...	Quiz7
123456787	70	90	...	80	76	...	70
123456788	91	85	...	73	90	...	80
123456789	60	71	...	78	88	...	73

(a) Create a simulated dataset in R¹ called **gradebook** that represents a possible gradebook in the basic format as Table 1:

- All columns are numeric.
- Each row of the gradebook should contain all measurements for a student.
- Each column should contain scores for one assignment.
- The last column should be “Quiz7.”
- The number of homework and quizzes is fixed. However, the number of students may vary while grading.

The simulated **gradebook** should contain the grades for 100 students and scores (out of 100) for 5 homework and 7 quizzes. Set the seed for simulating your data with your UID.

- (b) Write R code in an R Markdown file to randomly replace 10% of the values in Homework1, Homework5, and Quiz3 with NA, respectively. Then, use `is.na()` in conjunction with `sum()` to display the results. Note that the number of rows and the locations of the NA values may vary during grading.
- (c) Write a function **messy_impute()** that imputes missing values in the **gradebook**. Please also present your algorithm (or flowchart) to answer this question in the RMD file.

Note:

- Assume the format of the gradebook is fixed (five homework and seven quizzes), but NA values may occur in any assignments except for UID.
- The **messy_impute()** function should have at least three arguments and ...:
 - data: a **tibble** object contains the gradebook as specified in the example.
 - center: an argument should be a character object indicating the impute functions (mean, median, and min). The default center is “min”.
 - margin: an argument should be an integer (1 or 2) indicating either the function imputes the missing values across columns/assignments (1) or rows/students (2). If choosing option (2), the function should process homework and quizzes separately.
- The function should return the imputed tibble. In addition, the function should include error-handling to ensure the function works properly.
- **Toy example with three homework assignments:**

To impute the NA value across columns with mean, we compute $\lfloor \frac{60+23}{2} \rfloor = 41$ to replace the NA. If do it across rows, the function should compute $\lfloor \frac{52+88+51+54}{4} \rfloor = 61$ to replace the NA.

Note: The objective of this example is to illustrate the imputation process. Please generate and utilize the data as instructed in (a).

¹please check how to use `runif()`.

UID	Homework1	Homework2	Homework3
111111111	41.00	27.00	52.00
222222222	43.00	94.00	88.00
333333333	2.00	79.00	51.00
444444444	46.00	45.00	54.00
555555555	60.00	23.00	NA

- (d) Here are some suggested test cases but not limited to while grading. Please select one student whose quiz or homework has undergone imputation to showcase your function. Print only what you want to demonstrate in the R markdown file. **Don't print the entire gradebook.**
- `messy_impute(gradebook, "mean", 1)`
 - `messy_impute(gradebook, "median", 2)`
 - `messy_impute(gradebook, "mean", 1, trim = 0.25)`
- (e) Write R code using the main functions in the `tidyr` package to convert the **gradebook** into the tidy format. Name this object **gradebook_tidy**. You may directly write code in the R markdown file to answer this question.
- (f) Write a function **tidy_impute()** that imputes missing values in **gradebook_tidy** object. The **tidy_impute()** function should have the same arguments as in the **messy_impute()** function. You should return an imputed **gradebook_tidy** object. Please also present your algorithm or flowchart to answer this question in the R markdown file.
- Note:** You will not get credits for this function if convert **gradebook_tidy** object to be a messy format or reuse your *messy_impute()* in any steps of your **tidy_impute()**.
- (g) Please use the cases you select from (d) to demonstrate your function in the R markdown file. Print only what you want to demonstrate. **Don't print the entire gradebook.**

2: Short Answers

The formatting takes 10% of your homework grade. The accuracy takes 15% of your grade. The total is 25% for this question.

- (a) Please find three examples in which data might naturally be presented in a messy (non-tidy) way. For each example, include the context, observations, and variables that would be recorded.
- (b) You may search online to answer this question, but you **MUST** cite your sources. **Please don't make up data on your own.**
- (c) Get modest samples from each dataset you found and save them as CSV files in your homework folder. Each file should contain a minimum of 10 rows and 3 columns, with a maximum limit of 100 rows and 100 columns.
- (d) For each of the three examples in (a), describe how the data might be better presented in a tidy way. You have to use tidyverse functions to reorganize the small sample datasets in (c) into tidy format and show your results on your RMD file.